

MATH40002 - Analysis I

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Introduction

In the introductory module (MATH40001), we constructed the number systems $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$. Real analysis focuses on the real numbers, $\mathbb{R}$, formalising many properties about functions that we know from calculus. For example, an intuitive notion of a continuous function is that you can draw the function on a piece of paper without taking your pen off the page. We make concepts like these precise in this module.

Chapter 1

Countability

1.1 Countably infinite sets

Before starting this chapter (and, in general, this module), it is imperative that you are familiar with the logic and sets part of the introductory module, MATH40001.

Definition 1.1. A set A is countably infinite if there exists a bijection $f : \mathbb{N} \rightarrow A$.

Intuitively, this says that you can list the elements of A as $a_1, a_2, \dots$ using only the natural numbers. We say that a set is *countable* if it is finite or countably infinite. Note we will take $\mathbb{N} = \{1, 2, \dots\}$, unless stated otherwise.

Note that if there is a bijection between a given pair of sets A and B , then A is countable if and only if B is countable. This extends the natural idea of finite sets having the same number of elements if and only if there is a bijection between the sets. Formally, we define two sets having the same *cardinality* as follows:

Definition 1.2. Let A, B be sets. We say that A and B have the same cardinality if there exists a bijection $f : A \rightarrow B$.

The idea is that even in infinite settings, if we know a property about the 'size' of one set A , and there exists a bijection with another set B , then B shares this notion of 'size'. For example, as we mentioned, if such a bijection exists between two sets A, B , then A is countable if and only if B is countable. This gives rise to an equivalence relation, which we define as cardinality.

Exercise. Show that the relation $\sim$ on $\mathcal{P}(\mathcal{U})$ (where $\mathcal{U}$ is the universal set of interest) defined by $A \sim B \iff \exists f : A \rightarrow B$ such that f is a bijection, is an equivalence relation.

Theorem 1.3. If $A \subseteq \mathbb{N}$ is an infinite set, then A is countably infinite.

Proof. We construct a function $f : \mathbb{N} \rightarrow A$ inductively: since A is nonempty, and $A \subseteq \mathbb{N}$, by well-ordering of the natural numbers we can pick $a_1 = \min A$, and let $f(1) = a_1$. Assume that, for $k \geq 1$, $a_1, \dots, a_k$ have been chosen and $f(i) = a_i$ for $1 \leq i \leq k$. Since A is infinite, $A \setminus \{a_1, \dots, a_k\}$ is nonempty. Pick $a_{k+1} = \min(A \setminus \{a_1, \dots, a_k\})$, and let $f(k+1) = a_{k+1}$. Thus by construction f is strictly increasing. We claim that f is a bijection.

- Injective: this holds because by construction f is strictly increasing. We saw in the introductory module that strictly increasing functions are injective.
- Surjective: suppose for contradiction that f is not surjective. Then there exists $a \in A$ such that for all $n \in \mathbb{N}$, $f(n) \neq a$. Consider $A \setminus \{a\}$.

□

Lemma 1.4. *Let S be an infinite set. There exists an injection $f : \mathbb{N} \rightarrow S$.*

Proof. We construct a function $f : \mathbb{N} \rightarrow S$ inductively: since S is infinite, it is nonempty, so we can pick $s_1 \in S$ and define $f(1) := s_1$. Suppose for $k \in \mathbb{N}$, $s_1, \dots, s_k$ have been chosen and we have defined $f(i) = s_i$ for $1 \leq i \leq k$. Since S is infinite, $S \setminus \{s_1, \dots, s_k\}$ is nonempty, so there is $s_{k+1} \in S \setminus \{s_1, \dots, s_k\}$. Define $f(k+1) := s_{k+1}$. By construction, if $i < j$ are natural numbers, then $s_j \in S \setminus \{s_1, \dots, s_{j-1}\}$ so $s_j \neq s_i$. So f is an injection. □

As a result, we can always *embed* the natural numbers into any infinite set via an injection $f : \mathbb{N} \rightarrow S$ (where S is any infinite set). In this way, the natural numbers can be thought of as 'the smallest type of infinity'.

A not so surprising fact which we can prove immediately is that the set of integers, $\mathbb{Z}$, is countably infinite:

Theorem 1.5. *$\mathbb{Z}$ is countably infinite.*

Proof. Define $f : \mathbb{N} \rightarrow \mathbb{Z}$ by

$$f(n) = \begin{cases} n/2, & \text{if } n \text{ even} \\ -(n+1)/2, & \text{if } n \text{ odd} \end{cases}$$

Then f is injective: suppose $f(a) = f(b)$. If a and b are of different parity then $f(a)$ and $f(b)$ are of different sign, so cannot be equal. So this case cannot happen. If a and b are even, then $a/2 = b/2$ so $a = b$. If a and b are odd, then $-(a+1)/2 = -(b+1)/2$ so $a = b$. So f is injective. Also, f is surjective: let $m \in \mathbb{Z}$ be arbitrary. If $m < 0$ then $f(-2m-1) = m$, and if $m \geq 0$ then $f(2m) = m$. Thus f is a bijection and $\mathbb{Z}$ is countable. □

Remark. This proof used a useful trick called interleaving, where we listed the integers as $0, -1, 1, -2, 2, \dots$, interleaving the negative and positive integers to create the desired bijection between $\mathbb{N}$ and $\mathbb{Z}$.

A less trivial fact is that the rational numbers, $\mathbb{Q}$, is countable. This may be hard to see until observing the proof of the following fact:

Theorem 1.6. *$\mathbb{N}^2$ is countably infinite.*

Proof. The idea is illustrated in the above diagram. Note that the listing goes

$$(0, 0), (0, 1), (1, 0), (2, 0), (1, 1), (0, 2), \dots$$

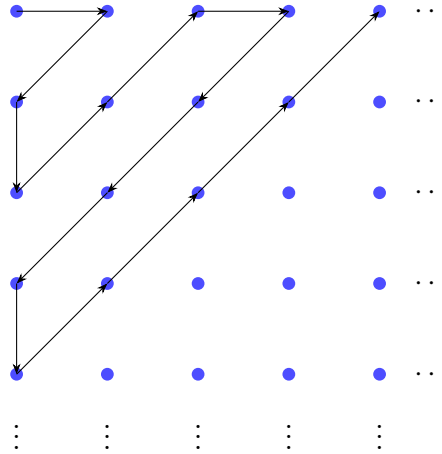


Figure 1.1: The diagonal listing of $\mathbb{N}^2$, showing it is countably infinite.

Notice that along each diagonal, $a + b$ is constant for all pairs (a, b) in that diagonal. This, along with the triangular number formula, provides the motivation for how to construct a bijection between $\mathbb{N}^2$ and $\mathbb{N}$.

Explicitly, define $f : \mathbb{N}^2 \rightarrow \mathbb{N}$ by $f(a, b) = \frac{(a+b)(a+b+1)}{2} + b$. This is a bijection (exercise) so $\mathbb{N}^2$ is countable. Note that here we took $\mathbb{N} = \{0, 1, 2, \dots\}$. $\square$

As a consequence, we get:

Theorem 1.7. $\mathbb{Q}$ is countably infinite.

Proof. First we use Theorem 1.6 to show that $\mathbb{Q}_{\geq 0} := \{x \in \mathbb{Q} : x \geq 0\}$ is countably infinite. Let

$$S = \{(a, b) \in \mathbb{N}^2 : b \geq 1, \gcd(a, b) = 1\}.$$

Then $f : S \rightarrow \mathbb{Q}_{\geq 0}$, defined by $f(a, b) = a/b$ is a bijection because we know that all rational numbers in $\mathbb{Q}_{\geq 0}$ take the unique form a/b where $b \geq 1$ and $\gcd(a, b) = 1$. Since $\mathbb{N}^2$ is countably infinite by Theorem 1.6, $S \subseteq \mathbb{N}^2$ is also countably infinite, so $\mathbb{Q}_{\geq 0}$ is countably infinite.

It follows (exercise) that $\mathbb{Q}$ is countably infinite, using the same idea (interleaving) as the proof that $\mathbb{Z}$ is countably infinite. $\square$

Exercise. Finish the proof of Theorem 1.7. (Hint: use the fact we proved above that $\mathbb{Q}_{\geq 0}$ is countably infinite, so there exists a bijection $h : \mathbb{N} \rightarrow \mathbb{Q}_{\geq 0}$).

The following lemma is a useful way of proving that a given set S is countably infinite without explicitly constructing a bijection $f : \mathbb{N} \rightarrow S$.

Lemma 1.8 (A criterion for countability). *Let S be an infinite set.*

1. S is countably infinite if and only if there exists an injection $\iota : S \rightarrow \mathbb{N}$.
2. S is countably infinite if and only if there exists a surjection $g : \mathbb{N} \rightarrow S$.

Proof. a $\square$

The idea for item (1) above is that if you can *embed* your set into the natural numbers via an injection $\iota : S \rightarrow \mathbb{N}$, then it must be countably infinite, as $\iota(S) \subseteq \mathbb{N}$, and we have shown infinite subsets of $\mathbb{N}$ are countable, so since ι is an injection, S has the same cardinality as $\iota(S)$, (i.e. $S \sim \iota(S)$), so S is countably infinite. The idea for item (2) is similar: if you can use the natural numbers to list all the elements of your infinite set S , then even though this surjective map $g : \mathbb{N} \rightarrow S$ may not be injective, we have nonetheless listed all the elements of S using only the natural numbers, so S must be countably infinite also.

A simple example using Lemma 1.8 is that finite products of countable sets are countable (note that we already have the tools to prove this without Lemma 1.8 by using Theorem 1.6, which is a more elementary way of proving this fact via an explicit bijection).

Before looking at the example, try and think of properties of the integers that we can use to construct an injection $\iota : A \times B \rightarrow \mathbb{N}$.

Example 1.9. If A, B are countably infinite sets, then $A \times B$ is countably infinite.

Proof. Recall from the introductory module that we define

$$A \times B := \{(a, b) : a \in A, b \in B\}.$$

Since A and B are countably infinite, let $f : \mathbb{N} \rightarrow A$, $g : \mathbb{N} \rightarrow B$ be bijections. Therefore, we can write $A = \{a_1, a_2, \dots\}$ and $B = \{b_1, b_2, \dots\}$. Define $\iota : A \times B \rightarrow \mathbb{N}$ by $\iota(a_i, b_j) = 2^i 3^j$. By the Fundamental Theorem of Arithmetic from the introductory module, ι is an injection because prime factorisations are unique, so by Lemma 1.8, $A \times B$ is countably infinite. $\square$

This trick of mapping each element of the set to a prime factorisation is useful to keep in mind, as it is a very neat way of proving countability for sets.

1.2 Uncountable sets

Not all sets are countably infinite:

Definition 1.10. An infinite set is called *uncountable* if it is not countably infinite.

For example, the set of real numbers $\mathbb{R}$ is uncountable. To show this, we first prove:

Lemma 1.11. The set $(0, 1) := \{x \in \mathbb{R} : 0 < x < 1\}$ is uncountable.

Proof. The main technique that we use in arguing why a given set is uncountable is *Cantor's diagonal argument*, which goes like this. Suppose, for contradiction, that $(0, 1)$ is countably infinite. Then there exists a bijection $f : \mathbb{N} \rightarrow (0, 1)$ (note here we will let $\mathbb{N} = \{1, 2, \dots\}$ for aesthetics; the argument does not change). For each $n \in \mathbb{N}$, we write $f(n) = x_n = 0.a_{1n}a_{2n}\dots$, writing each x_n in decimal form, excluding tails of the digit 9 (for example, $0.8999\dots = 0.9$). We now get a listing of

the numbers in $(0, 1)$ which looks like:

$$\begin{aligned} f(1) &= 0.a_{11}a_{12}a_{13}a_{14}\dots \\ f(2) &= 0.a_{21}a_{22}a_{23}a_{24}\dots \\ f(3) &= 0.a_{31}a_{32}a_{33}a_{34}\dots \\ f(4) &= 0.a_{41}a_{42}a_{43}a_{44}\dots \\ &\vdots \end{aligned}$$

The idea is that we look at the *diagonal* elements, a_{ii} for $i \geq 1$, and construct a number $x \in (0, 1)$ such that for all $i \geq 1$, the i^{th} decimal place of x differs from the i^{th} decimal place of $f(i)$, thereby constructing a number $x \in (0, 1)$ that is not in the enumeration f of the interval $(0, 1)$, which is a contradiction to f being surjective. We construct $x \in (0, 1)$ as follows: for $j \geq 1$, let b_j be the j^{th} decimal place of x , then we define

$$b_j = \begin{cases} 2 & \text{if } a_{jj} = 1 \\ 1 & \text{if } a_{jj} \neq 1 \end{cases}$$

Then by construction, $b_i \neq a_{ii}$ for each $i \geq 1$, so in particular (by uniqueness of decimal representations under our restriction to decimal expansions with no digit 9 tails), we have $x \neq f(i)$ for all $i \geq 1$, so $x \in (0, 1)$ is not in the image of f , a contradiction to f being surjective onto $(0, 1)$. Therefore, $(0, 1)$ is uncountable. $\square$

Theorem 1.12. $\mathbb{R}$ is uncountable.

Proof. Follows from Lemma 1.11 (exercise). $\square$

Exercise. Find a bijection $f : (0, 1) \rightarrow \mathbb{R}$, and deduce that $\mathbb{R}$ is an uncountable set.

Chapter 2

Sequences and Series

2.1 Sequences

Definition 2.1. A real sequence is a function $f : \mathbb{N} \rightarrow \mathbb{R}$.

In general, $\mathbb{R}$ can be replaced by a metric space X , for example the complex numbers $\mathbb{C}$ which we will see later. It is therefore important to distinguish *real* sequences. For notation, we will write $(a_n)_{n=1}^{\infty}$ to denote a sequence $a_1, a_2, \dots$, and sometimes this will be shortened to (a_n) , or just a_n .

2.1.1 Convergence of sequences

Intuitively, a real sequence converges to a limit $L \in \mathbb{R}$ if we can make the terms of our sequence, a_n , arbitrarily close to L , whenever we go far enough out into the sequence. Clearly, this does not suffice as a mathematical definition, so instead we define convergence of a real sequence (a_n) as follows.

Definition 2.2 (Convergence of a sequence in $\mathbb{R}$). A real sequence (a_n) converges to $L \in \mathbb{R}$ if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, |a_n - L| < \varepsilon.$$

For notation, if Definition 2.2 holds for a sequence (a_n) , we will write $\lim_{n \rightarrow \infty} a_n = L$, which we will sometimes shorten to $a_n \rightarrow L$ (as $n \rightarrow \infty$) or $\lim a_n = L$, if from the context it is clear we are talking about a sequence, and not, say, a limit of a function $f(x)$ (which, as we will see, does not have to be the limit as $x \rightarrow \infty$).

Example 2.3. Use Definition 2.2 to show that the sequence (a_n) defined by $a_n = 1/n$ for each $n \in \mathbb{N}$ converges to 0.

Proof. Let $\varepsilon > 0$ be arbitrary. Choose $N \in \mathbb{N}$ such that $N > 1/\varepsilon$ (such an N exists by the Archimedean property of the real numbers from the introductory module). Then, given $n \geq N$, we have

$$|a_n - 0| = 1/n \leq 1/N < \varepsilon,$$

so $a_n \rightarrow 0$. □

Definition 2.4. A sequence *diverges* if it does not converge to any $L \in \mathbb{R}$.

Definition 2.5. A real sequence (a_n) is bounded if there exists $M > 0$ such that for all $n \in \mathbb{N}$, $|a_n| \leq M$.

Remark. Note that a real sequence can diverge but still be bounded (see below).

Example 2.6. Use Definition 2.2 to show that the sequence $a_n := (-1)^n$ diverges.

Proof. Suppose for contradiction that $a_n \rightarrow L \in \mathbb{R}$. Let $N \in \mathbb{N}$ be such that for all $n \geq N$,

$$|(-1)^n - L| < 1/2.$$

But also, by the triangle inequality,

$$2 = |(-1)^{2N} - (-1)^{2N+1}| \leq |(-1)^{2N} - L| + |L - (-1)^{2N+1}| < 1/2 + 1/2 = 1,$$

which is a contradiction. Therefore, (a_n) diverges. $\square$

2.1.2 Basic results on convergent sequences

Unsurprisingly, if a sequence converges to a limit, then it must be unique:

Theorem 2.7. *Limits of convergent sequences are unique.*

Proof. Suppose $a_n \rightarrow L$ and $a_n \rightarrow M$, and let $\varepsilon > 0$ be arbitrary. We wish to conclude that $L = M$. By convergence of $a_n \rightarrow L$, pick $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|a_n - L| < \varepsilon/2$. By convergence of $a_n \rightarrow M$ let $N_2 \in \mathbb{N}$ be such that for all $n \geq N_2$, $|a_n - M| < \varepsilon/2$. Now, let $n := \max\{N_1, N_2\}$. By the triangle inequality, we have

$$|L - M| = |L - a_n + a_n - M| \leq |a_n - L| + |a_n - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore, since ε was arbitrary, $L = M$ (to see this, if $L \neq M$ then one can pick $\varepsilon = |L - M|/2$ and get a contradiction). $\square$

Theorem 2.8 (Order Limit Theorem). *Suppose (a_n) and (b_n) are two real sequences such that $a_n \geq b_n$ for all $n \in \mathbb{N}$. Let $a_n \rightarrow a$ and $b_n \rightarrow b$. Then $a \geq b$.*

Proof. Suppose for contradiction that $a < b$. Pick $\varepsilon = \frac{b-a}{2}$. Since $a_n \rightarrow a$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|a_n - a| < \varepsilon$, so in particular for $n \geq N_1$,

$$a_n < a + \varepsilon = a + \frac{b-a}{2} = \frac{a+b}{2},$$

and since $b_n \rightarrow b$ there exists $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$, $|b_n - b| < \varepsilon$, so in particular for $n \geq N_2$,

$$b_n > b - \varepsilon = b - \frac{b-a}{2} = \frac{a+b}{2}.$$

Therefore, choosing $n := \max\{N_1, N_2\}$ it follows that $b_n > a_n$, which is a contradiction. $\square$

Corollary 2.9. Suppose (a_n) is a real sequence with $a_n \rightarrow a$ such that $B \leq a_n \leq C$ for all $n \in \mathbb{N}$. Then $B \leq a \leq C$.

Proof. Apply Theorem 2.8 using constant sequences $b_n = B$ and $c_n = C$ for all $n \in \mathbb{N}$. $\square$

Remark. If $B < a_n < C$ for all $n \in \mathbb{N}$ in the above corollary, it is not necessarily true that $B < a < C$, but we do still get the non-strict inequalities $B \leq a \leq C$. Limits only preserve non-strict inequalities.

Theorem 2.10 (Squeeze Theorem). Let (a_n) , (b_n) and (c_n) be real sequences such that for all $n \in \mathbb{N}$, $a_n \leq b_n \leq c_n$. Suppose that $a_n \rightarrow L$ and $c_n \rightarrow L$. Then $b_n \rightarrow L$.

Proof. Let $\varepsilon > 0$ be arbitrary. Since $a_n \rightarrow L$, we can pick $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|a_n - L| < \varepsilon$. Since $c_n \rightarrow L$, we can also pick $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$, $|c_n - L| < \varepsilon$. Let $N := \max\{N_1, N_2\}$. Then given $n \geq N$, we have $a_n, c_n \in (L - \varepsilon, L + \varepsilon)$, i.e.

$$L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon,$$

so $|b_n - L| < \varepsilon$, so $b_n \rightarrow L$. $\square$

Example 2.11. Let $a_n = \frac{\sin n}{n}$. Then for all $n \in \mathbb{N}$ we have

$$-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n},$$

since $|\sin n| \leq 1$ for all $n \in \mathbb{N}$. Therefore, since $1/n \rightarrow 0$, $a_n \rightarrow 0$ by the Squeeze Theorem $a_n \rightarrow 0$.

Theorem 2.12. Convergent sequences are bounded.

Proof. Let (a_n) be a convergent sequence, and let $L \in \mathbb{R}$ be such that $\lim a_n = L$. Let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|a_n - L| < 1$. In other words, for all $n \geq N$ we have

$$a_n \in (L - 1, L + 1).$$

Now, pick $M := \max\{|a_1|, |a_2|, \dots, |a_{N-1}|, |L| + 1\}$. Then by construction, $|a_n| \leq M$ for all $n \in \mathbb{N}$, so (a_n) is bounded. $\square$

Remark. The idea for the above proof is that we can control (a_n) to be close to L (say, within a distance of 1 from L), past some point $N \in \mathbb{N}$, and before that, there are only finitely many terms in the sequence, so we can just take the maximum of a finite number of terms $|a_1|, \dots, |a_{N-1}|$, and then include $|L| + 1$ to complete the bound.

Similarly to the above observation, changing a finite number of terms in a convergent sequence does not affect its convergence. When dealing with convergence of sequences, we only care about the behaviour of the *tail* of the sequence. This motivates the following definition:

Definition 2.13. A sequence (a_n) *eventually* has a property P if there exists $N \in \mathbb{N}$ such that for all $n \geq N$, a_n has property P .

Remark. Modifying the hypothesis of a theorem which requires a property of a sequence to hold for all $n \in \mathbb{N}$ with *eventually* does not weaken the theorem; they are equivalent.

Example 2.14. Let the property P be 'constant'. The sequence $a_1 = 1, a_2 = 2$ and $a_n = 0$ for all $n \geq 3$ is *eventually* constant.

Exercise. Prove that if an integer sequence (a_n) (that is, $a_n \in \mathbb{Z}$ for all $n \in \mathbb{N}$) converges, then it is eventually constant.

Definition 2.15. Let (a_n) be a real sequence.

- $a_n \rightarrow \infty$ (as $n \rightarrow \infty$) if for every $R > 0$, a_n is eventually greater than R .
- $a_n \rightarrow -\infty$ (as $n \rightarrow \infty$) if for every $R < 0$, a_n is eventually less than R .

Example 2.16.

1. $a_n = n^2 \rightarrow \infty$. Indeed, let $R > 0$, and take $n > \sqrt{R}$. Then

$$a_n = n^2 > (\sqrt{R})^2 = R.$$

2. Define

$$a_n = \begin{cases} 0 & \text{if } n \text{ odd} \\ n & \text{if } n \text{ even} \end{cases}$$

This is an example of a sequence which does not diverge to ∞ , but is unbounded (as an exercise, use Definition 2.15 to prove this directly).

Theorem 2.17. Let (a_n) and (b_n) be real sequences.

1. If $a_n \rightarrow \infty$ and b_n is bounded below, then $a_n + b_n \rightarrow \infty$.
2. If $a_n \rightarrow \infty$ and $\lambda \in \mathbb{R}$, then

$$\lambda a_n \rightarrow \begin{cases} \infty, & \lambda > 0, \\ 0, & \lambda = 0, \\ -\infty, & \lambda < 0. \end{cases}$$

3. If (a_n) is bounded and $b_n \rightarrow \infty$ with $b_n \neq 0$ for all $n \in \mathbb{N}$ then $\frac{a_n}{b_n} \rightarrow 0$.
4. If $a_n \rightarrow \infty$ and there exists $B > 0$ such that eventually $b_n > B$, then $a_n b_n \rightarrow \infty$.

Proof. 1. Suppose $a_n \rightarrow \infty$ and let $M > 0$ be such that $b_n \geq M$ for all $n \in \mathbb{N}$. Let $R > 0$ be arbitrary. Since $a_n \rightarrow \infty$, let $N \in \mathbb{N}$ be such that for all $n \geq N$, $a_n > R - M$. Then given $n \geq N$, we have

$$a_n + b_n \geq a_n + M > R - M + M = R,$$

so $a_n + b_n \rightarrow \infty$.

2. Suppose $a_n \rightarrow \infty$, let $\lambda \in \mathbb{R}$. If $\lambda > 0$, let $R > 0$ be arbitrary. Then since $a_n \rightarrow \infty$ let $N \in \mathbb{N}$ be such that for all $n \geq N$, $a_n > R/\lambda$. Then for $n \geq N$, we have $\lambda a_n > \lambda(R/\lambda) = R$, so $\lambda a_n \rightarrow \infty$. If $\lambda = 0$, then λa_n is the constant zero sequence, so converges to 0. If $\lambda < 0$, let $R < 0$ be arbitrary. Since $a_n \rightarrow \infty$ let $N \in \mathbb{N}$ be such that for all $n \geq N$, $a_n > R/\lambda$. Then for $n \geq N$ we have $\lambda a_n < \lambda(R/\lambda) = R$, so $a_n \rightarrow -\infty$.

3. Let $M > 0$ be such that $|a_n| \leq M$ for all $n \in \mathbb{N}$ and suppose $b_n \rightarrow \infty$ with $b_n \neq 0$ for all $n \in \mathbb{N}$. Let $\varepsilon > 0$ be arbitrary; since $b_n \rightarrow \infty$, we can pick $N \in \mathbb{N}$ such that for all $n \geq N$, $b_n > M/\varepsilon$. Therefore, for $n \geq N$, we have

$$\left| \frac{a_n}{b_n} \right| \leq \frac{M}{b_n} < \frac{M}{M/\varepsilon} = \varepsilon,$$

so $\frac{a_n}{b_n} \rightarrow 0$.

4. Suppose $a_n \rightarrow \infty$ and let $B > 0$; suppose that $N_1 \in \mathbb{N}$ is such that for all $n \geq N_1$, $b_n > B$. Now, let $R > 0$ be arbitrary. Since $a_n \rightarrow \infty$, let $N_2 \in \mathbb{N}$ be such that for all $n \geq N_2$, $a_n > R/B$. Let $N := \max\{N_1, N_2\}$. Then given $n \geq N$ we have

$$a_n b_n > a_n B > \left(\frac{R}{B} \right) B = R,$$

so $a_n b_n \rightarrow \infty$. □

Theorem 2.18 (Growth comparisons).

1. Let $m \in \mathbb{R}$ with $m \neq 0$. Then

$$\lim_{n \rightarrow \infty} n^m = \begin{cases} \infty, & m > 0, \\ 0, & m < 0. \end{cases}$$

2. Let $\lambda \in \mathbb{R}$ with $\lambda \neq 1$. Then

$$\lim_{n \rightarrow \infty} \lambda^n = \begin{cases} \infty, & \lambda > 1, \\ 0, & |\lambda| < 1. \end{cases}$$

If $\lambda \leq -1$, then λ^n diverges.

3. Let $m > 0$ and $\lambda > 1$. Then $\lim_{n \rightarrow \infty} \frac{\lambda^n}{n^m} = \infty$ (“exponentials beat powers of n ”).

4. Let $m > 0$. Then $\lim_{n \rightarrow \infty} \frac{\log n}{n^m} = 0$ (“powers of n beat logs”).

Proof. 1. We will prove the case for $m > 0$; the case for $m < 0$ follows immediately by item 3 in Theorem 2.17. Let $R > 0$ be arbitrary, and pick $N \in \mathbb{N}$ such that $N > R^{1/m}$. Then for $n \geq N$ we have $n^m \geq N^m > (R^{1/m})^m = R$, since $x \mapsto x^m$ is an increasing function for $m > 0$, $x > 0$.

2. Suppose $\lambda > 1$. Then we have by Bernoulli's inequality, for $n \in \mathbb{N}$,

$$\lambda^n = (1 + (\lambda - 1))^n \geq 1 + (\lambda - 1)n \rightarrow \infty,$$

since $\lambda - 1 > 0$.

Suppose $0 < |\lambda| < 1$. Then there exists $\mu > 1$ such that $|\lambda| = 1/\mu$. By the result for $\lambda > 1$ and item 3 in Theorem 2.17, we have that $|\lambda|^n = 1/\mu^n \rightarrow 0$, since $\mu^n \rightarrow \infty$ by the $\lambda > 1$ case we just proved. Because $|\lambda^n| \rightarrow 0$, it follows that $\lambda^n \rightarrow 0$. Suppose $\lambda \leq -1$. If $\lambda = -1$, the sequence is $(-1)^n$, which we have already shown diverges in an example. If $\lambda < -1$, then $|\lambda| > 1$, so by the first part, $|\lambda|^n \rightarrow \infty$. Since the absolute value of the sequence diverges to infinity, λ^n cannot converge to a finite limit, and therefore diverges.

3. Let $\lambda > 1$. Then we can write $\lambda = 1 + \delta$ for some real number $\delta > 0$. Let $k \in \mathbb{N}$ be an integer strictly greater than m . For any $n \geq k$, we can expand λ^n using the Binomial Theorem:

$$\lambda^n = (1 + \delta)^n = \sum_{j=0}^n \binom{n}{j} \delta^j.$$

Since $\delta > 0$, every term in this sum is positive. We can establish a lower bound by dropping all terms except the k -th term:

$$\lambda^n \geq \binom{n}{k} \delta^k = \frac{n(n-1) \cdots (n-k+1)}{k!} \delta^k.$$

For $n \geq 2k$, the first k terms in the factorial expansion $(n, n-1, \dots, n-k+1)$ are all greater than or equal to $n/2$. Thus, we can further bound the binomial coefficient:

$$\binom{n}{k} \geq \frac{(n/2)^k}{k!} = \frac{n^k}{2^k k!}.$$

Substituting this back into our ratio, for $n \geq 2k$:

$$\frac{\lambda^n}{n^m} \geq \frac{n^k}{n^m} \frac{\delta^k}{2^k k!} = n^{k-m} \left(\frac{\delta^k}{2^k k!} \right).$$

Notice that $\frac{\delta^k}{2^k k!}$ is a strictly positive constant independent of n . Because we chose $k > m$, the exponent $k - m$ is strictly positive. Therefore, n^{k-m} diverges to ∞ as $n \rightarrow \infty$ by item 1 in this theorem. Since $\frac{\lambda^n}{n^m}$ is bounded below by a sequence diverging to ∞ , we conclude that $\lim_{n \rightarrow \infty} \frac{\lambda^n}{n^m} = \infty$.

4. We will prove this when we rigorously define the exp and log functions in Chapter 6. Furthermore, it is much easier to prove this once we have built some more theory. $\square$

Changing a finite number of terms in a convergent sequence does not affect convergence:

Lemma 2.19. *Let $K \in \mathbb{N}$ be arbitrary. Then $(a_n)_{n=1}^{\infty}$ is a convergent sequence with limit $L \in \mathbb{R}$ if and only if $(a_{n+K})_{n=1}^{\infty}$ is a convergent sequence, with limit $L \in \mathbb{R}$.*

Proof. Suppose (a_n) converges to $L \in \mathbb{R}$. Pick $N \in \mathbb{N}$ such that for all $n \geq N$, $|a_n - L| < \varepsilon$. Since $K \geq 0$, given $n \geq N$, $n + K \geq n \geq N$, so $|a_{n+K} - L| < \varepsilon$.

Conversely, suppose (a_{n+K}) converges to $L \in \mathbb{R}$. Let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|a_{n+K} - L| < \varepsilon$. Choose $N' := N + K$. Given $n \geq N'$, we have $n \geq N + K$, so for $n \geq N'$, $|a_n - L| < \varepsilon$. $\square$

2.1.3 Algebra of Limits

There is a useful theorem that allows us to avoid ε proofs and use convergence results that we have already proven:

Theorem 2.20 (Algebra of Limits). *Let (a_n) , and (b_n) be real sequences, and suppose $a_n \rightarrow a \in \mathbb{R}$, and $b_n \rightarrow b \in \mathbb{R}$. Then*

1. $a_n + b_n \rightarrow a + b$.
2. $a_n b_n \rightarrow ab$.
3. For all $\lambda \in \mathbb{R}$, $\lambda a_n \rightarrow \lambda a$.
4. If $b_n \neq 0$ for all $n \in \mathbb{N}$ and $b \neq 0$, then $1/b_n \rightarrow 1/b$.

Proof. (1) and (3) are good exercises to practice using Definition 2.2.

2. Case 1: without loss of generality, suppose that $a \neq 0$ and $b = 0$; let $\varepsilon > 0$ be arbitrary. By Theorem 2.12, there exists $M > 0$ such that for all $n \in \mathbb{N}$, $|a_n| \leq M$. By convergence of $b_n \rightarrow 0$, let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|b_n| < \varepsilon/M$. Then for $n \geq N$, we have

$$|a_n b_n| = |a_n| |b_n| \leq M |b_n| < M(\varepsilon/M) = \varepsilon,$$

so $a_n b_n \rightarrow 0$.

Case 2: Suppose that $a \neq 0$ and $b \neq 0$; let $\varepsilon > 0$ be arbitrary. By Theorem 2.12 let $M > 0$ be such that for all $n \in \mathbb{N}$, $|a_n| \leq M$. By convergence of $a_n \rightarrow a$, let $N_1 \in \mathbb{N}$ be such that for all $n \geq N_1$, $|a_n - a| < \varepsilon/2|b|$. Also, by convergence of $b_n \rightarrow b$, let $N_2 \in \mathbb{N}$ be such that for all $n \geq N_2$, $|b_n - b| < \varepsilon/2M$. Choose $N := \max\{N_1, N_2\}$. Then for all $n \geq N$, we have

$$\begin{aligned} |a_n b_n - ab| &= |a_n b_n - a_n b + a_n b - ab| \\ &\leq |a_n b_n - a_n b| + |a_n b - ab| \\ &= |a_n| |b_n - b| + |b| |a_n - a| \\ &\leq M |b_n - b| + |b| |a_n - a| \\ &< M \left(\frac{\varepsilon}{2M} \right) + |b| \left(\frac{\varepsilon}{2|b|} \right) \\ &= \varepsilon, \end{aligned}$$

so $a_n b_n \rightarrow ab$.

4. Let $\varepsilon > 0$ be arbitrary. Let $N_1 \in \mathbb{N}$ be such that for all $n \geq N_1$, $|b_n - b| < |b|/2$, so by the triangle inequality we have, for $n \geq N_1$,

$$|b| = |b - b_n + b_n| \leq |b_n - b| + |b_n| < \frac{|b|}{2} + |b_n| \implies |b_n| > \frac{|b|}{2} > 0.$$

Now, let $N_2 \in \mathbb{N}$ be such that for all $n \geq N_2$, $|b_n - b| < \frac{\varepsilon|b|^2}{2}$. Choose $N := \max\{N_1, N_2\}$. For all $n \geq N$, we have

$$\left| \frac{1}{b_n} - \frac{1}{b} \right| = \frac{|b_n - b|}{|b||b_n|} < \frac{|b_n - b|}{|b|^2/2} < \frac{\varepsilon|b|^2/2}{|b|^2/2} = \varepsilon,$$

so $1/b_n \rightarrow 1/b$. □

Remark. Using items 2 and 4 above, it follows that if $b \neq 0$ and $b_n \neq 0$ for all $n \in \mathbb{N}$, and $a_n \rightarrow a$, $b_n \rightarrow b$, then $a_n/b_n \rightarrow a/b$.

Example 2.21. 1. Let $a_n = \frac{n^2+4n+1}{2n^2+1}$. For $n \in \mathbb{N}$ we have

$$a_n = \frac{1 + 4/n + 1/n^2}{2 + 1/n^2}.$$

Now, using 2 in the Algebra of Limits (Theorem 2.20), $1/n^2 \rightarrow 0$ since we have shown $1/n \rightarrow 0$, and using 1, 3 and 4 in the Algebra of Limits we get $\lim a_n = \frac{1}{2}$.

2. Applying (2) inductively, since $1/n \rightarrow 0$ we get that for any $m \in \mathbb{N}$, $1/n^m \rightarrow 0$.
3. Let $a_n = \sqrt{n^2 + n} - n$. The standard trick is to multiply by the conjugate to convert the $\infty - \infty$ form into a situation where we can apply the Algebra of Limits; for $n \in \mathbb{N}$ we have

$$\begin{aligned} a_n &= \left(\sqrt{n^2 + n} - n \right) \frac{\sqrt{n^2 + n} + n}{\sqrt{n^2 + n} + n} = \frac{n}{\sqrt{n^2 + n} + n} \\ &= \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} \rightarrow \frac{1}{2}, \end{aligned}$$

by the Algebra of Limits, since $1/n \rightarrow 0$.

2.1.4 Monotone Convergence Theorem

Definition 2.22. Let (a_n) be a real sequence.

1. We say that a_n is *increasing* if for all $n \in \mathbb{N}$, $a_{n+1} \geq a_n$. If the inequality is strict for all $n \in \mathbb{N}$, we say that a_n is *strictly increasing*.
2. We say that a_n is *decreasing* if for all $n \in \mathbb{N}$, $a_{n+1} \leq a_n$. If the inequality is strict for all $n \in \mathbb{N}$, we say that a_n is *strictly decreasing*.
3. A sequence a_n is *monotonic* if it is increasing or decreasing.

Example 2.23. The sequence $a_n = 1/n$ is strictly decreasing and converges to 0, the sequence $a_n = 1 - 1/n$ is strictly increasing and converges to 1, the sequence $a_n = n$ is strictly increasing and diverges to ∞ , and the sequence $a_n = (-1)^n n$ is neither strictly increasing nor strictly decreasing, and diverges. Constant sequences are both increasing and decreasing.

We now state and prove an important theorem about monotonic and bounded real sequences.

Theorem 2.24 (Monotone Convergence Theorem). *Let (a_n) be a bounded and monotone real sequence. Then (a_n) is convergent.*

Proof. We will prove the case where a_n is increasing. The case where a_n is decreasing is similar, and is left as an exercise.

Consider the set $S := \{a_n : n \in \mathbb{N}\}$. S is nonempty since $a_1 \in S$, and since a_n is bounded, the set S is bounded. By the completeness of $\mathbb{R}$, $s := \sup S$ exists. We claim that $a_n \rightarrow s$.

Let $\varepsilon > 0$. By definition of supremum, there exists $N \in \mathbb{N}$ such that $s - \varepsilon < a_N \leq s$. But since s is an upper bound, and a_n is increasing, we therefore have

$$s - \varepsilon < a_N \leq a_{N+1} \leq \cdots \leq s < s + \varepsilon,$$

i.e. for all $n \geq N$, we have $|a_n - s| < \varepsilon$, so $a_n \rightarrow s$. □

Remark. This result is useful because it allows us to deduce convergence of a sequence without knowing what the limit is.

Example 2.25. Define a sequence (a_n) by $a_{n+1} = \sqrt{2 + a_n}$ for $n \in \mathbb{N}$ with $a_1 = \sqrt{2}$. We claim that (a_n) converges.

Proof. By induction, $0 < a_n < 2$ for all $n \in \mathbb{N}$; indeed, given $k \in \mathbb{N}$, if $a_k < 2$ then $a_{k+1} = \sqrt{2 + a_k} < \sqrt{2 + 2} = 2$, and the base case $a_1 = \sqrt{2} < 2$. So $0 < a_n < 2$ for all $n \in \mathbb{N}$. Therefore, a_n is bounded. Also, the sequence is strictly increasing: for $n \in \mathbb{N}$ we have

$$a_n = \sqrt{a_n a_n} < \sqrt{2a_n} = \sqrt{a_n + a_n} < \sqrt{a_n + 2} = a_{n+1},$$

so by Theorem 2.24, (a_n) is convergent.

In fact, with recurrence relations, we can often find the limit as well, once we know that it converges: this is because $\lim a_{n+1} = \lim a_n = L$ for some $L \in \mathbb{R}$, which must satisfy, by the Algebra of Limits (Theorem 2.20), $L = \sqrt{2 + L}$, i.e.

$$L^2 - L - 2 = 0 \implies (L - 2)(L + 1) = 0,$$

but since $0 < a_n < 2$ for all $n \in \mathbb{N}$, $0 \leq L \leq 2$ by Theorem 2.8 (Order Limit Theorem), so $L = 2$. □

2.1.5 Subsequences

Definition 2.26. Let (a_n) be a sequence, and let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a strictly increasing function. A *subsequence* of (a_n) is a sequence defined by $b_n := a_{f(n)}$ for $n \in \mathbb{N}$.

Remark. In practice, if $\{n_k : k \in \mathbb{N}\}$ is any strictly increasing set of indices such that $n_1 < n_2 < n_3 < \cdots$, the corresponding subsequence of $(a_n)_{n=1}^\infty$ is $(a_{n_k})_{k=1}^\infty$.

Example 2.27. Let $a_n = n$ for $n \in \mathbb{N}$. Then $a_{2n} = 2n$ is a subsequence of a_n . Let $a_n = 1/n$ for $n \in \mathbb{N}$. Then $a_p = 1/p$ for p prime is a subsequence of a_n . However, $(1/3, 1/2, 1, 1/4, 1/5, \dots)$ is not a subsequence of a_n (we changed the ordering of the terms).

Lemma 2.28. *Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a strictly increasing function. Then for all $k \in \mathbb{N}$, $f(k) \geq k$.*

Proof. The proof is by induction on k . For the base case, clearly $f(1) \geq 1$ since the codomain is $\mathbb{N}$, and 1 is the smallest natural number. Given $m \in \mathbb{N}$, suppose that $f(m) \geq m$. Since f is strictly increasing, $f(m+1) > f(m)$ i.e. $f(m+1) \geq f(m) + 1$, since $f(m+1) - f(m) \geq 1$ (smallest distance between distinct integers is 1), so

$$f(m+1) \geq f(m) + 1 \geq m + 1,$$

by the induction hypothesis, so $f(k) \geq k$ for all $k \in \mathbb{N}$. $\square$

Theorem 2.29. *Let (a_n) be a real sequence that converges to $L \in \mathbb{R}$. Then any subsequence of (a_n) converges to L .*

Proof. Let (a_{n_k}) be an arbitrary subsequence of (a_n) , and let $\varepsilon > 0$. Since $a_n \rightarrow L$, pick $N \in \mathbb{N}$ such that for all $n \in \mathbb{N}$, $|a_n - L| < \varepsilon$. Let $k \geq N$. Then by Lemma 2.28, we have $n_k \geq k \geq N$, so $|a_{n_k} - L| < \varepsilon$, so $a_{n_k} \rightarrow L$ as $k \rightarrow \infty$. $\square$

Remark. In contrapositive form, Theorem 2.29 is very useful for showing that a sequence diverges. You only need to find two subsequences that converge to different limits, to prove that a sequence diverges.

Example 2.30. We now get a much easier proof for the divergence of $a_n = (-1)^n$. We have $a_{2n} = 1$ for all $n \in \mathbb{N}$ and $a_{2n-1} = -1$ for all $n \in \mathbb{N}$, so by Theorem 2.29 (the contrapositive), a_n diverges.

Theorem 2.31. *Any real sequence has a monotonic subsequence.*

Proof. Let (a_n) be any real sequence. We consider the set of *peak indices* in the sequence, $S := \{n \in \mathbb{N} : a_n \geq a_m \forall m \geq n\}$. If $k \in S$, we say that a_k is *peak*.

Case 1: S is finite (or empty). Let $m \in \mathbb{N}$ be such that $m > n$ for all $n \in S$ (note that this vacuously includes the case where S is empty). Since $k \notin S$ for all $k \geq m$, in particular for all $k \geq m$, a_k is not *peak*. We therefore construct an increasing subsequence of (a_n) by induction. Define $n_1 := m$. Since a_{n_1} is not *peak*, there exists $n_2 > n_1$ such that $a_{n_2} > a_{n_1}$. Given $k \in \mathbb{N}$, suppose that $a_{n_1} < a_{n_2} < \dots < a_{n_k}$ have been chosen. Since a_{n_k} is not *peak*, there exists $n_{k+1} > n_k$ such that $a_{n_{k+1}} > a_{n_k}$. By construction, this is a (strictly) increasing subsequence of (a_n) .

Case 2: S is infinite. By the well-ordering of the natural numbers, we may pick $n_1 := \min S$. Given $k \in \mathbb{N}$, suppose that $n_1 < n_2 < \dots < n_k$ have been chosen such that $n_i \in S$ for all $1 \leq i \leq k$. Since S is infinite, $S \setminus \{n_1, \dots, n_k\}$ is nonempty, so we may pick $n_{k+1} := \min(S \setminus \{n_1, \dots, n_k\})$. By definition of the set S , for all $k \in \mathbb{N}$, $a_{n_k} \geq a_{n_{k+1}}$, since by construction a_{n_k} is *peak* for all $k \in \mathbb{N}$, so (a_n) is a decreasing subsequence of (a_n) . $\square$

Theorem 2.32 (Bolzano-Weierstraß Theorem). *Any bounded real sequence has a convergent subsequence.*

Proof. Let (a_n) be a bounded sequence. By Theorem 2.31, (a_n) has a monotone subsequence (a_{n_k}) , which is also bounded since (a_n) is bounded, so by the Monotone Convergence Theorem (Theorem 2.24), (a_{n_k}) is convergent. $\square$

Example 2.33. The sequence $a_n = \sin(n)$ has a convergent subsequence since $|a_n| \leq 1$ for all $n \in \mathbb{N}$. Finding such a subsequence is harder and is beyond the scope of this course.

2.1.6 Cauchy sequences

Convergent sequences, we have seen, get closer and closer to some limit L . What can we say about sequences whose terms get arbitrary close together, past some fixed point $N \in \mathbb{N}$? First, we define precisely what this means:

Definition 2.34. A real sequence (a_n) is *Cauchy* if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall m, n \geq N, |a_m - a_n| < \varepsilon.$$

A useful fact we can immediately prove (in a similar way to Theorem 2.12) is that Cauchy sequences are bounded.

Lemma 2.35. *Cauchy sequences are bounded.*

Proof. Exercise. $\square$

In fact, in $\mathbb{R}$, the concepts of Cauchy sequences and convergent sequences are equivalent in the following sense:

Theorem 2.36 (Cauchy criterion for convergence in $\mathbb{R}$). *Let (a_n) be a real sequence. Then*

$$(a_n) \text{ is convergent} \iff (a_n) \text{ is Cauchy.}$$

Proof. Suppose that (a_n) is a convergent real sequence with limit $L \in \mathbb{R}$, and let $\varepsilon > 0$ be arbitrary. Let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|a_n - L| < \frac{\varepsilon}{2}$. Then given $m, n \geq N$, we have by the triangle inequality

$$|a_m - a_n| = |(a_m - L) + (L - a_n)| \leq |a_m - L| + |a_n - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

so (a_n) is Cauchy.

Conversely, suppose that (a_n) is Cauchy. By Lemma 2.35, (a_n) is bounded. By the Bolzano-Weierstraß Theorem, there exists a convergent subsequence of (a_{n_k}) of (a_n) with limit $L \in \mathbb{R}$. We claim that $a_n \rightarrow L$. Let $\varepsilon > 0$ be arbitrary. Since $\lim_{k \rightarrow \infty} a_{n_k} = L$, we can let $N_1 \in \mathbb{N}$ be such that for all $n_j \geq N_1$, $|a_{n_j} - L| < \frac{\varepsilon}{2}$. Also, (a_n) is Cauchy, so we can let $N_2 \in \mathbb{N}$ be such that for all $m, n \geq N_2$, $|a_m - a_n| < \frac{\varepsilon}{2}$. Now, set $N := \max\{N_1, N_2\}$ and let $n_i \in \mathbb{N}$ be large enough so that $n_i \geq N$. Then given $n \geq N$ we have $|a_n - a_{n_i}| < \frac{\varepsilon}{2}$ and so by the triangle inequality, we get

$$|a_n - L| \leq |a_n - a_{n_i}| + |a_{n_i} - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

so $a_n \rightarrow L$ and (a_n) is convergent. $\square$

The key idea for the proof above of Cauchy in $\mathbb{R} \implies$ convergent in $\mathbb{R}$ for a given sequence is that by the Bolzano-Weierstraß theorem, there exists a convergent subsequence of (a_n) , but at the same time, since (a_n) is Cauchy, the terms in the sequence are getting arbitrarily close together, while the convergent subsequence terms are getting arbitrarily close to a limit $L \in \mathbb{R}$, so we can use this to force any sufficiently far out term in the sequence to be arbitrarily close to L as well.

Remark. It is important to note that the above equivalence, in particular Cauchy $\implies$ convergent does not hold for all sets (however, convergence implies Cauchy in any metric space). It relies on the fact that $\mathbb{R}$ is a *complete* set, that is, it has the least upper bound property, which means that, intuitively, all sequences in $\mathbb{R}$ that should converge (*Cauchy* sequences) actually have a limit within that space. For example, $\mathbb{Q}$ is not a complete set (indeed, $\mathbb{R}$ is the completion of $\mathbb{Q}$, as we saw when constructing $\mathbb{R}$ in the introductory module), so the fact that Cauchy sequences are convergent does not hold in $\mathbb{Q}$. For example, $(1 + 1/n)^n \rightarrow e \notin \mathbb{Q}$. We will prove the irrationality of e later, and we may define $e := \sup\{(1 + 1/n)^n : n \in \mathbb{N}\}$; then by the Monotone Convergence Theorem (Theorem 2.24), $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$ (note that there are more practical ways of defining e and the $\exp : \mathbb{R} \rightarrow (0, \infty)$ function, as we will see in Chapter 6).

Remark. In addition to the above comment about completeness, we notice that the Bolzano-Weierstraß Theorem was required to prove that Cauchy sequences are convergent. In fact, there is an equivalence between the following concepts for complete sets (in particular, $\mathbb{R}$ in our case):

- Least upper bound property: every non-empty subset of $\mathbb{R}$ that is bounded above has a supremum (least upper bound) in $\mathbb{R}$.
- Cauchy completeness: Every Cauchy sequence in $\mathbb{R}$ converges to a limit in $\mathbb{R}$.
- Bolzano-Weierstraß Theorem: Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.
- Monotone convergence Theorem: Every bounded, monotonic sequence in $\mathbb{R}$ converges.
- Nested interval property: Any sequence of nested, closed, and bounded intervals $[a_n, b_n]$ has a nonempty intersection.

Exercise. Prove that the above concepts are equivalent.

2.1.7 Complex sequences

As hinted at earlier, sequences need not be real. An interesting and rich area of mathematics is analysis of the complex numbers, $\mathbb{C}$, which we will see more of in second year. Here, we build some theory on complex sequences (and later, complex series and complex power series). There will be no rigorous construction of $\mathbb{C}$ (however, it is not too bad now that we have constructed $\mathbb{R}$ because, in fact, we can construct $\mathbb{C}$ from $\mathbb{R}^2$ with some special operations, namely the usual complex addition and complex multiplication). We will assume all the usual properties of the complex numbers, in particular that $\mathbb{C}$ is a field with the usual complex addition and complex multiplication.

Definition 2.37. A complex sequence is a function $f : \mathbb{N} \rightarrow \mathbb{C}$.

Again, practically, we write (z_n) or just z_n to denote a complex sequence. For a complex sequence (z_n) it is natural to associate two real sequences $(x_n), (y_n)$ with (z_n) such that for each $n \in \mathbb{N}$, $z_n = x_n + iy_n$, i.e. $x_n = \Re(z_n)$ and $y_n = \Im(z_n)$, where $\Re$ and $\Im$ denote the real and imaginary parts of a complex number, respectively. It is therefore not too hard to prove facts about complex sequences using our results about real sequences.

Definition 2.38 (Convergence of a sequence in $\mathbb{C}$). A complex sequence (z_n) converges to $z \in \mathbb{C}$ if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, |z_n - z| < \varepsilon.$$

Remark. Note that the modulus $|\cdot|$ is now the *complex* modulus defined by $|z| := \sqrt{x^2 + y^2}$ where $z = x + iy$.

First, we prove a convergence criterion for complex sequences, in relation to its real and imaginary parts:

Theorem 2.39 (Parts criterion for convergence in $\mathbb{C}$). Let (z_n) be a complex sequence, and define real sequences $(x_n), (y_n)$ by $x_n := \Re(z_n)$ and $y_n := \Im(z_n)$ for all $n \in \mathbb{N}$. Let $z \in \mathbb{C}$ and write $z = x + iy$ for some $x, y \in \mathbb{R}$. Then

$$z_n \rightarrow z \iff x_n \rightarrow x \text{ and } y_n \rightarrow y.$$

Proof. Suppose $z_n \rightarrow z = x + iy$, and let $\varepsilon > 0$ be arbitrary. Let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|z_n - z| < \varepsilon$. Given $n \geq N$, we have

$$|x_n - x| = \sqrt{(x_n - x)^2} \leq \sqrt{(x_n - x)^2 + (y_n - y)^2} = |z_n - z| < \varepsilon,$$

and similarly

$$|y_n - y| \leq \sqrt{(x_n - x)^2 + (y_n - y)^2} = |z_n - z| < \varepsilon,$$

so $x_n \rightarrow x$ and $y_n \rightarrow y$.

Conversely, suppose $x_n \rightarrow x$ and $y_n \rightarrow y$. Let $N_1 \in \mathbb{N}$ be such that for all $n \geq N_1$, $|x_n - x| < \frac{\varepsilon}{\sqrt{2}}$, and let $N_2 \in \mathbb{N}$ be such that for all $n \geq N_2$, $|y_n - y| < \frac{\varepsilon}{\sqrt{2}}$. Let $N := \max\{N_1, N_2\}$. Given $n \geq N$ we have

$$|z_n - z| = \sqrt{(x_n - x)^2 + (y_n - y)^2} < \sqrt{\frac{\varepsilon^2}{2} + \frac{\varepsilon^2}{2}} = \varepsilon,$$

so $z_n \rightarrow z$. □

Example 2.40. 1. Let $z_n = \frac{1}{n} + in$. This does not converge because $\Im(z_n) = n \rightarrow \infty$. Geometrically, since $\Re(z_n) = \frac{1}{n} \rightarrow 0$, this sequence gets arbitrarily close to the imaginary axis in the complex plane, but diverges from the real axis.

2. For $k \in \mathbb{N}$, define the sequence $z_n = e^{\frac{2n\pi i}{k}}$; this does not converge (cycles around a regular k -gon centred at $(0,0)$). Exercise: what are the possible subsequential limits of this sequence? (i.e. if (z_{n_k}) is a subsequence of (z_n) , what are the possible limits?)

3. Fix $z_1 \in \mathbb{C}$ and for $n \in \mathbb{N}$ define $z_{n+1} = \frac{z_n + |z_n|}{2}$. We claim that this sequence converges. First, for $n \in \mathbb{N}$ we write $z_n = x_n + iy_n$, so we have

$$z_{n+1} = \frac{x_n + iy_n + |z_n|}{2},$$

and $|z_n|$ is real for all $n \in \mathbb{N}$, so we have $x_{n+1} = \Re(z_{n+1}) = \frac{x_n + |z_n|}{2}$ and $y_{n+1} = \Im(z_{n+1}) = \frac{y_n}{2}$. Therefore, by induction $y_{n+1} = \frac{y_1}{2^n}$ so $y_n \rightarrow 0$. Furthermore, we know that

$$|z_n| = \sqrt{x_n^2 + y_n^2} \geq \sqrt{x_n^2} = |x_n|,$$

so $x_{n+1} \geq x_n$ for all $n \in \mathbb{N}$. Also, by construction of (z_n) , $|z_n| \leq |z_1|$ for all $n \in \mathbb{N}$ so $|x_n| \leq |z_n| \leq |z_0|$ for all $n \in \mathbb{N}$. Therefore, (x_n) is an increasing sequence and is bounded above, so by the Monotone Convergence Theorem, x_n converges to some $L \in \mathbb{R}$. Therefore, by Theorem 2.39, $z_n \rightarrow L$.

Definition 2.41. A complex sequence (z_n) is bounded if there exists $M \in \mathbb{R}_{>0}$ such that $|z_n| \leq M$ for all $n \in \mathbb{N}$.

Theorem 2.42. Let (z_n) be a complex sequence. If (z_n) converges, then (z_n) is bounded.

Proof. Using the parts criterion for convergence in $\mathbb{C}$ (Theorem 2.39), $x_n := \Re(z_n)$ and $y_n := \Im(z_n)$ are convergent and so by Theorem 2.12 (x_n) and (y_n) are bounded. Let $M_1 \in \mathbb{R}_{>0}$ be such that $|x_n| \leq M_1$ for all $n \in \mathbb{N}$ and let $M_2 \in \mathbb{R}_{>0}$ be such that $|y_n| \leq M_2$ for all $n \in \mathbb{N}$. Choose $M := \max\{M_1, M_2\} \in \mathbb{R}_{>0}$; then $|x_n| \leq M$ and $|y_n| \leq M$ for all $n \in \mathbb{N}$, so for all $n \in \mathbb{N}$ we have

$$|z_n| = \sqrt{x_n^2 + y_n^2} \leq \sqrt{2}M,$$

so (z_n) is bounded. □

Theorem 2.43 (Bolzano-Weierstraß Theorem in $\mathbb{C}$). Any bounded complex sequence has a convergent subsequence.

Proof. Let (z_n) be a bounded complex sequence. Put $x_n := \Re(z_n)$ and $y_n := \Im(z_n)$. Then for all $n \in \mathbb{N}$, $|x_n| = \sqrt{x_n^2} \leq \sqrt{x_n^2 + y_n^2} = |z_n|$ and $|y_n| \leq |z_n|$, so (x_n) and (y_n) are bounded. By the Bolzano-Weierstraß Theorem in $\mathbb{R}$ (Theorem 2.32), there exists a convergent subsequence (x_{n_k}) of (x_n) . Consider the subsequence (y_{n_k}) of (y_n) . By the Bolzano-Weierstraß Theorem in $\mathbb{R}$, there exists a convergent subsequence $(y_{n_{k_j}})$ of (y_{n_k}) . Furthermore, by Theorem 2.29, $(x_{n_{k_j}})$ is a convergent subsequence of (x_n) , so by the parts criterion for convergence in $\mathbb{C}$, $(z_{n_{k_j}})$ is a convergent subsequence of (z_n) . □

Definition 2.44. A complex sequence is *Cauchy* if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall m, n \geq N, |z_m - z_n| < \varepsilon.$$

Theorem 2.45 (Cauchy criterion for convergence in $\mathbb{C}$). *Let (z_n) be a complex sequence and let $x_n := \Re(z_n)$, $y_n := \Im(z_n)$ for all $n \in \mathbb{N}$. Then the following are equivalent:*

1. (z_n) is Cauchy in $\mathbb{C}$.
2. (x_n) and (y_n) are Cauchy in $\mathbb{R}$.
3. (z_n) converges in $\mathbb{C}$.

Proof. $1 \Rightarrow 2$: For $m, n \in \mathbb{N}$ we have $|x_m - x_n| = |\Re(z_m - z_n)| \leq |z_m - z_n|$, so (x_n) is Cauchy in $\mathbb{R}$, and similarly for (y_n) (same argument, just use imaginary part).

$2 \Rightarrow 3$: By the Cauchy Criterion for Convergence in $\mathbb{R}$ (Theorem 2.36), $x_n \rightarrow x$ and $y_n \rightarrow y$ for some $x, y \in \mathbb{R}$, so by Theorem 2.39, $z_n \rightarrow x + iy = z$.

$3 \Rightarrow 1$: Suppose $z_n \rightarrow z$ for some $z \in \mathbb{C}$, and let $\varepsilon > 0$ be arbitrary. Since $z_n \rightarrow z$, we can let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|z_n - z| < \frac{\varepsilon}{2}$. Then given $m, n \geq N$ we have, by the triangle inequality,

$$|z_m - z_n| \leq |z_m - z| + |z_n - z| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

so (z_n) is Cauchy in $\mathbb{C}$. □

2.2 Series

2.2.1 Partial sums

Definition 2.46. A complex sequence (a_n) is called *summable* (or *convergent as a series*) if the sequence of partial sums $s_n := \sum_{i=1}^n a_i$ converges to some limit s as $n \rightarrow \infty$. We then write $\sum_{n=1}^{\infty} a_n = s$.

Some remarks:

- Note that complex sequences include real sequences.
- A series is fundamentally just a special type of sequence—the sequence of its partial sums. All theorems about series are ultimately just theorems about sequences applied to S_N .
- A series is fundamentally a *limit of finite sums*, rather than an “infinite sum”. It is dangerous to treat a series as an infinite form of addition because infinite series do not always obey the standard rules of finite arithmetic (such as commutativity and associativity), which we will see when we look at divergent series and conditionally convergent series. Because it is a limit, the *order* in which we take the limit matters, which explains why rearranging the terms of conditionally convergent series can alter the final result.
- Rather than saying a sequence (a_n) is summable, we may also say that $\sum_{n=1}^{\infty} a_n$ converges, or sometimes if from the context it is clear that we are talking about series, we may just write $\sum a_n$ converges.

- When talking about divergent series, we often abuse notation and write $\sum_{n=1}^{\infty} a_n$ diverges, even though writing $\sum_{n=1}^{\infty} a_n$ does not make sense in this case (it is not a real number, we just use it symbolically in this case).

Example 2.47. The geometric series $\sum_{n=1}^{\infty} (1/2)^n$ has partial sums $S_N = 1 - (1/2)^N$ (exercise: show this), which converge to 1.

Theorem 2.48 (Algebra of Limits for Series). *If $\sum a_n$ and $\sum b_n$ are converge, and $\lambda \in \mathbb{C}$, then $\sum(a_n + b_n)$ and $\sum \lambda a_n$ are also summable, with:*

$$\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n \quad \text{and} \quad \sum_{n=1}^{\infty} \lambda a_n = \lambda \sum_{n=1}^{\infty} a_n.$$

Proof. This is a simple application of the Algebra of Limits (Theorem 2.20) on the sequences $s_n := \sum_{i=1}^n a_i \rightarrow s$ and $t_n := \sum_{i=1}^n b_i$. In particular, set $u_n = \sum_{i=1}^n (a_i + b_i)$. Then we have

$$u_n = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i = s_n + t_n \rightarrow s + t,$$

i.e. $\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$, as required. Furthermore, set $v_n = \sum_{i=1}^n \lambda a_i$. Then we have

$$v_n = \lambda s_n \rightarrow \lambda s,$$

by the Algebra of Limits for sequences again, i.e. $\sum_{n=1}^{\infty} \lambda a_n = \lambda \sum_{n=1}^{\infty} a_n$. $\square$

Lemma 2.49 (Shift summability lemma). *A complex sequence (a_n) is summable if and only if the shifted sequence (a_{n+K}) is summable for any fixed $K \in \mathbb{N}$.*

Proof. Define $b_n := \sum_{i=K+1}^{n+K} a_i = \sum_{i=1}^n a_{i+K}$. Then we have

$$b_n = \sum_{i=1}^{n+K} a_i - \sum_{i=1}^K a_i = \sum_{i=1}^{n+K} a_i - C,$$

where $C := \sum_{i=1}^K a_i$ is a constant. Now, suppose (a_n) is summable. Therefore, $\lim_{n \rightarrow \infty} \sum_{i=1}^{n+K} a_i$ exists and we have that $\lim_{n \rightarrow \infty} b_n$ exists with $\sum_{n=1}^{\infty} a_{n+K} = \sum_{n=1}^{\infty} a_n - C$, using the Algebra of Limits. Conversely, suppose that (a_{n+K}) is summable, i.e. $\lim_{n \rightarrow \infty} b_n$ exists. Therefore, from the above, $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_{n+K} + C$ using the Algebra of Limits. $\square$

Remark. Convergence is strictly a property of the “tail” of a sequence. Adding, removing, or changing the first million terms of a series changes its final sum, but it cannot change *whether* the series converges or diverges.

2.2.2 Convergence tests for series

Theorem 2.50 (Geometric series). *The series $\sum_{n=0}^{\infty} r^n$ converges if and only if $|r| < 1$. Moreover, if $|r| < 1$ then*

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

Proof. Let $r \in \mathbb{R}$ and put $s_n = \sum_{i=0}^n r^i$. By an exercise, $s_n = \frac{1-r^{n+1}}{1-r}$. By item 2 in Theorem 2.18, $\lim_{n \rightarrow \infty} r^n$ exists if and only if $|r| < 1$, so s_n converges if and only if $|r| < 1$. Moreover, if $|r| < 1$ then by item 2 in Theorem 2.18 we have $r^n \rightarrow 0$ as $n \rightarrow \infty$, therefore

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r},$$

as desired. $\square$

Theorem 2.51 (Necessary condition for summability). *If the complex sequence (a_n) is summable, then $\lim_{n \rightarrow \infty} a_n = 0$.*

Proof. Suppose (a_n) is a summable complex sequence. Then $s_n := \sum_{i=1}^n a_i$ converges. Therefore, (s_n) is Cauchy, so in particular given $\varepsilon > 0$ we can pick $N \in \mathbb{N}$ such that for all $m, n \geq N$, we have $|s_m - s_n| < \varepsilon$. Therefore, letting $n \geq N$ be arbitrary and picking $m = n + 1$, we have

$$|a_n| = |s_{n+1} - s_n| < \varepsilon,$$

so $a_n \rightarrow 0$. $\square$

Example 2.52. 1. Let $a_n = (1 + \frac{1}{n})^n$. Then $a_n \rightarrow e \geq 1 > 0$, so $\sum_{n=1}^{\infty} a_n$ diverges by Theorem 2.51.

2. Sometimes, $a_n \rightarrow 0$ but the series still diverges: let $s_n = \sum_{i=1}^n \log(\frac{i+1}{i})$. Then

$$s_n = \sum_{i=1}^n (\log(i+1) - \log(i)) = \log(n+1) \rightarrow \infty,$$

so $\sum_{n=1}^{\infty} \log(\frac{n+1}{n})$ diverges, even though $\log(1 + \frac{1}{i}) \rightarrow 0$ as $i \rightarrow \infty$.

Remark. Note that this is a necessary, but *not sufficient* condition for series convergence. The classic counterexample is the harmonic series $\sum \frac{1}{n}$. Even though $\frac{1}{n} \rightarrow 0$, the series diverges to infinity because the terms do not shrink *fast enough*. We will prove this now.

Theorem 2.53 (Divergence of the Harmonic Series). *The series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.*

Proof. Let $s_n = \sum_{i=1}^n \frac{1}{i}$ denote the partial sums of $\frac{1}{n}$. We consider s_{2^n} :

$$\begin{aligned} s_{2^n} &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots + \underbrace{\left(\frac{1}{2^{n-1}+1} + \cdots + \frac{1}{2^n}\right)}_{2^{n-1} \text{ terms}} \\ &\geq 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^n} + \cdots + \frac{1}{2^n}\right) \\ &= 1 + \frac{1}{2} + 2\left(\frac{1}{4}\right) + 4\left(\frac{1}{2^3}\right) + \cdots + 2^{n-1}\left(\frac{1}{2^n}\right) \\ &= 1 + \frac{n}{2} \rightarrow \infty, \end{aligned}$$

so $s_{2^n} \rightarrow \infty$ as $n \rightarrow \infty$. Therefore, given $R > 0$, let $n \in \mathbb{N}$ be such that $s_{2^n} > R$. Since (s_n) is an increasing sequence, for $m \geq 2^n$ we have $s_m \geq s_{2^n} > R$, so $s_n \rightarrow \infty$ as $n \rightarrow \infty$. $\square$

Therefore, despite the incredibly slow pace ($\mathcal{O}(\log n)$), this sequence of partial sums eventually surpasses every positive real number.

In fact, the argument used in the previous result can be generalised:

Theorem 2.54 (Cauchy Condensation test). *Let (a_n) be a decreasing sequence of non-negative real numbers. Then*

$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \sum_{n=0}^{\infty} 2^n a_{2^n} \text{ converges}.$$

Proof. Let $s_n := \sum_{i=1}^n a_i$ and $t_n := \sum_{j=0}^n 2^j a_{2^j}$. We will show that for all $k \in \mathbb{N}$,

$$\frac{t_k}{2} \leq s_{2^k} \leq t_k,$$

which will give us the desired result. Indeed, given $k \in \mathbb{N}$ we have

$$\begin{aligned} s_{2^k} &= a_1 + a_2 + (a_3 + a_4) + (a_5 + a_6 + a_7 + a_8) + \cdots + (a_{2^{k-1}+1} + \cdots + a_{2^k}) \\ &\geq \frac{1}{2}a_1 + a_2 + 2a_4 + 4a_8 + \cdots + 2^{k-1}a_{2^k} \\ &= \frac{1}{2} \left(a_1 + 2a_2 + 4a_4 + 8a_8 + \cdots + 2^k a_{2^k} \right) \\ &= \frac{t_k}{2}, \end{aligned}$$

and

$$\begin{aligned} s_{2^k} &= a_1 + (a_2 + a_3) + (a_4 + a_5 + a_6 + a_7) + \cdots + (a_{2^{k-1}} + \cdots + a_{2^k-1}) + a_{2^k} \\ &\leq a_1 + (a_2 + a_2) + (a_4 + a_4 + a_4 + a_4) + \cdots + (a_{2^{k-1}} + \cdots + a_{2^k-1}) + 2^k a_{2^k} \\ &= t_k, \end{aligned}$$

as desired. The result follows. $\square$

This is a powerful test and can be used to determine the convergence or divergence of a large class of infinite series. Indeed, a simple (but useful) corollary is:

Corollary 2.55 (*p*-series test). *The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.*

Proof. Put $a_n = \frac{1}{n^p}$ and let $t_n = \sum_{i=0}^n 2^i a_{2^i}$. Then we have, for $n \in \mathbb{N}$,

$$t_n = \sum_{i=0}^n 2^i \frac{1}{(2^i)^p} = \sum_{i=0}^n 2^{i(1-p)}.$$

This is a geometric sum, so by Theorem 2.50, t_n converges if and only if $2^{1-p} < 1$ i.e. if and only if $p > 1$. By the Cauchy Condensation test (Theorem 2.54), it follows that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$. $\square$

Exercise. Determine the values of p for which the series $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ converges.

Theorem 2.56 (Comparison test). *Let (a_n) and (b_n) be real sequences such that $0 \leq a_n \leq b_n$ for all n (or eventually).*

1. *If $\sum b_n$ is convergent, then $\sum a_n$ is convergent.*
2. *If $\sum a_n$ diverges, then $\sum b_n$ diverges.*

Proof. Let $s_n = \sum_{i=1}^n a_i$ and $t_n = \sum_{i=1}^n b_i$. Since $0 \leq a_i \leq b_i$ for all $i \in \mathbb{N}$, we can sum this inequality to get that $0 \leq s_n \leq t_n$ for all $n \in \mathbb{N}$. Furthermore, observe that since $a_i \geq 0$ for all $i \in \mathbb{N}$, (s_n) is an increasing sequence.

1. Suppose $\sum b_n$ is convergent. Then (t_n) is bounded, so there exists $M > 0$ such that for all $n \in \mathbb{N}$, $t_n \leq M$. Therefore, using the above observation, $s_n \leq t_n \leq M$, so (s_n) is bounded, and is increasing so by the Monotone Convergence Theorem (Theorem 2.24), (s_n) is convergent, i.e. $\sum a_n$ is convergent.
2. Suppose $\sum a_n$ diverges, i.e. (s_n) diverges. Then since $t_n \geq s_n \rightarrow \infty$, t_n diverges i.e. $\sum b_n$ diverges. $\square$

Remark. If item 1 in the above theorem holds, we say that “ $\sum a_n$ converges by comparison with $\sum b_n$ ” (and similarly for item 2).

Example 2.57. 1. For $n \in \mathbb{N}$ we have $0 \leq \frac{1}{n^2+1} \leq \frac{1}{n^2}$ and by the *p*-series test (Corollary 2.55), $\frac{1}{n^2}$ is summable, so $\frac{1}{n^2+1}$ is summable.

2. For $n \in \mathbb{N}$ we have $0 \leq \frac{2+\cos n}{n^3} \leq \frac{3}{n^3}$ since $|\cos n| \leq 1$ so since $\frac{1}{n^3}$ is summable by the *p*-series test, $\sum_{n=1}^{\infty} \frac{2+\cos n}{n^3}$ is convergent.

Definition 2.58 (Absolute and conditional convergence). A complex sequence (a_n) is *absolutely summable* if the sequence of absolute values $(|a_n|)$ is summable. If (a_n) is summable but not absolutely summable, it is called *conditionally summable*.

Remark. Absolute convergence means the series converges purely based on the rapidly decreasing magnitude of its terms, even in the worst-case scenario where every term is positive. Conditional convergence means the series only converges

because of cancellation between positive and negative terms. If you take the negative terms in your sequence, take the absolute value and construct a sequence of partial sums, and similarly construct a sequence of partial sums for the positive terms, absolute convergence means that both partial sums converge. It is therefore not surprising that rearranging terms arbitrarily in your sequence does not change the value of an absolutely convergent series (we will prove this later). However, rearranging terms in a conditionally convergent series.

Example 2.59. $\sum \frac{(-1)^n}{n^2}$ is absolutely convergent. $\sum \frac{(-1)^n}{n}$ is conditionally convergent (it converges, but its absolute value is the harmonic series which we saw diverges).

Theorem 2.60. *If a real sequence (a_n) is absolutely summable, then it is summable.*

Proof. Suppose $(|a_n|)$ is summable. For $n \in \mathbb{N}$ we have $0 \leq a_n + |a_n| \leq 2|a_n|$, and $(|a_n|)$ is summable, so $(2|a_n|)$ is summable, so by the Comparison test (Theorem 2.56), $(a_n + |a_n|)$ is summable. Therefore, $a_n = (a_n + |a_n|) - |a_n|$ is summable. $\square$

Theorem 2.61 (Alternating Series test). *Let (a_n) be a sequence of non-negative real numbers that is decreasing and $a_n \rightarrow 0$ as $n \rightarrow \infty$. Then the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ is convergent.*

Proof. Let (a_n) be a sequence such that $a_n \geq 0$, $a_{n+1} \leq a_n$ for all $n \in \mathbb{N}$, and $\lim_{n \rightarrow \infty} a_n = 0$.

Let s_n denote the n -th partial sum, $s_n = \sum_{k=1}^n (-1)^{k+1} a_k$. We will first examine the even partial sums, s_{2n} . We can group the terms of s_{2n} as follows:

$$s_{2n} = (a_1 - a_2) + (a_3 - a_4) + \cdots + (a_{2n-1} - a_{2n})$$

Because the sequence (a_n) is monotonically decreasing, we know that $a_{2k-1} - a_{2k} \geq 0$ for all k . Therefore, s_{2n} is a sum of non-negative terms, which implies that the sequence of even partial sums (s_{2n}) is monotonically increasing: $s_{2n} \leq s_{2(n+1)}$.

Alternatively, we can rewrite the grouping of s_{2n} by isolating the first term:

$$s_{2n} = a_1 - (a_2 - a_3) - (a_4 - a_5) - \cdots - (a_{2n-2} - a_{2n-1}) - a_{2n}$$

Again, because $a_k \geq a_{k+1}$, every term in parentheses is non-negative, and $a_{2n} > 0$. Thus, we are subtracting purely non-negative quantities from a_1 , which guarantees that $s_{2n} \leq a_1$ for all n .

Since (s_{2n}) is monotonically increasing and bounded above by a_1 , it must converge to some real number s by the Monotone Convergence Theorem. That is,

$$\lim_{n \rightarrow \infty} s_{2n} = s$$

Next, we examine the odd partial sums, s_{2n+1} . We can express the $(2n+1)$ -th partial sum purely in terms of the $2n$ -th partial sum and the next sequence term:

$$s_{2n+1} = s_{2n} + a_{2n+1}$$

Taking the limit of both sides as $n \rightarrow \infty$, and using the initial hypothesis that $\lim_{n \rightarrow \infty} a_n = 0$, we find, by the Algebra of Limits:

$$\lim_{n \rightarrow \infty} s_{2n+1} = \lim_{n \rightarrow \infty} (s_{2n} + a_{2n+1}) = \lim_{n \rightarrow \infty} s_{2n} + \lim_{n \rightarrow \infty} a_{2n+1} = s + 0 = s$$

Because the sequence of even partial sums (s_{2n}) and the sequence of odd partial sums (s_{2n+1}) both converge to the exact same limit s , the sequence of all partial sums (s_m) must also converge to s . Therefore, the alternating series converges. $\square$

Theorem 2.62 (Ratio test). *Let (a_n) be a real sequence of non-zero terms. Suppose that $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$.*

1. *If $L < 1$, the series $\sum a_n$ is absolutely summable.*
2. *If $L > 1$, the series diverges.*
3. *If $L = 1$, the test is inconclusive.*

Proof. Let (a_n) be a real sequence with $a_n > 0$ for all $n \in \mathbb{N}$, and suppose $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$.

Case 1: $L < 1$ (convergence). Choose $r \in \mathbb{R}$ such that $L < r < 1$. Because the limit of the ratio is $L < 1$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have:

$$\frac{a_{n+1}}{a_n} < r$$

This implies that $a_{n+1} < ra_n$ for all $n \geq N$. By induction, for any $k \geq 1$ we get:

$$a_{N+k} < ra_{N+k-1} < r^2 a_{N+k-2} < \cdots < r^k a_N$$

We are interested in the tail of the original series, $\sum_{k=1}^{\infty} a_{N+k}$. From the inequality above, we have:

$$\sum_{k=1}^{\infty} a_{N+k} < \sum_{k=1}^{\infty} r^k a_N = a_N \sum_{k=1}^{\infty} r^k$$

Since $0 < r < 1$, the geometric series $\sum r^k$ converges. By the Direct Comparison Test, the tail $\sum a_{N+k}$ converges, and therefore the entire series $\sum a_n$ converges (by the shift summability lemma).

Case 2: $L > 1$ (divergence). Choose a real number r such that $1 < r < L$. By the definition of the limit, there exists an integer $N \in \mathbb{N}$ such that for all $n \geq N$:

$$\frac{a_{n+1}}{a_n} > r > 1$$

This means $a_{n+1} > ra_n$ for all $n \geq N$. Since the sequence of terms (a_n) is strictly increasing and strictly positive for $n \geq N$, it is impossible for $\lim_{n \rightarrow \infty} a_n = 0$. By the n -th term divergence test (Theorem 2.51), the series $\sum a_n$ diverges. $\square$

Theorem 2.63 (Root test). *Let (a_n) be a real sequence. Suppose that*

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L.$$

Then

1. If $L < 1$, the series $\sum a_n$ is absolutely summable.
2. If $L > 1$, the series diverges.
3. If $L = 1$, the test is inconclusive.

Proof. Let (a_n) be a real sequence with $a_n \geq 0$ for all $n \in \mathbb{N}$, and suppose $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = L$.

Case 1: $L < 1$ (convergence). Choose $r \in \mathbb{R}$ such that $L < r < 1$. By the definition of the limit, there exists an integer $N \in \mathbb{N}$ such that for all $n \geq N$:

$$\sqrt[n]{a_n} < r$$

Raising both sides to the n -th power, since both sides are non-negative, yields:

$$a_n < r^n \quad \text{for all } n \geq N$$

Because $0 \leq r < 1$, the geometric series $\sum_{n=N}^{\infty} r^n$ converges. By the Direct Comparison Test, the tail of our series $\sum_{n=N}^{\infty} a_n$ converges, which means the entire series $\sum a_n$ converges.

Case 2: $L > 1$ (divergence). Since $L > 1$, there exists an integer $N \in \mathbb{N}$ such that for all $n \geq N$:

$$\sqrt[n]{a_n} > 1$$

Raising both sides to the n -th power gives:

$$a_n > 1^n = 1 \quad \text{for all } n \geq N$$

Because the terms of the series are eventually bounded below by 1, we know that $\lim_{n \rightarrow \infty} a_n \neq 0$. By the n -th term divergence test, the series $\sum a_n$ diverges. $\square$

Example 2.64. 1. The series $\sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges. Set $a_n = \frac{2^n}{n!}$ for $n \in \mathbb{N}$. Then we have

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{2^{n+1}/(n+1)!}{2^n/n!} = \frac{2}{n+1} \rightarrow 0 < 1,$$

so $\sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges by the Ratio test (Theorem 2.62).

2. The series $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges. Set $a_n = \frac{n}{2^n}$ for $n \in \mathbb{N}$. Then we have

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/2^{n+1}}{n/2^n} = \frac{1}{2} \frac{n+1}{n} \rightarrow \frac{1}{2} < 1,$$

so $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges by the Ratio test.

3. The series $\sum_{n=1}^{\infty} \left(\frac{4n^2-3n}{5n^2+1} \right)^n$ converges. Set $a_n = \left(\frac{4n^2-3n}{5n^2+1} \right)^n$ for $n \in \mathbb{N}$. Then we have

$$\sqrt[n]{a_n} = \frac{4n^2-3n}{5n^2+1} = \frac{4-\frac{3}{n}}{5+\frac{1}{n^2}} \rightarrow \frac{4}{5} < 1,$$

by the Algebra of Limits. Therefore, by the Root test (Theorem 2.63), $\sum_{n=1}^{\infty} \left(\frac{4n^2-3n}{5n^2+1} \right)^n$ converges.

Remark. Both the root and ratio tests examine whether the sequence in question can be bounded above ($L < 1$) or below ($L > 1$) by a convergent or divergent geometric series. If $L < 1$, it decays exponentially fast (like $(1/2)^n$), so it converges. If $L > 1$, it grows at least as fast as a divergent geometric series (intuitively), so it diverges.

Example 2.65. For $a_n = \frac{1}{n!}$, the ratio is $\left| \frac{1/(n+1)!}{1/n!} \right| = \frac{1}{n+1} \rightarrow 0 < 1$. Thus, factorials grow so fast that their reciprocals decay much faster than any geometric series, guaranteeing absolute convergence.

The following series test is useful for series with oscillating terms.

Theorem 2.66 (Dirichlet test). *Let (a_n) and (b_n) be sequences of real numbers such that:*

1. (b_n) is decreasing and $b_n \rightarrow 0$ as $n \rightarrow \infty$.
2. The partial sums of (a_n) are bounded, i.e., $\exists M > 0$ such that $|\sum_{i=1}^n a_i| \leq M$ for all $n \in \mathbb{N}$.

Then the series $\sum_{n=1}^{\infty} a_n b_n$ converges.

Proof. Let (a_n) and (b_n) be sequences of real numbers such that:

1. The partial sums $A_n = \sum_{i=1}^n a_i$ are bounded. That is, there exists a real number $M > 0$ such that $|A_n| \leq M$ for all $n \in \mathbb{N}$.
2. The sequence (b_n) is decreasing ($b_{n+1} \leq b_n$ for all $n \in \mathbb{N}$).
3. $\lim_{n \rightarrow \infty} b_n = 0$.

Let $S_n = \sum_{k=1}^n a_k b_k$. First, we rewrite S_n using summation by parts (also known as Abel's lemma). Since $a_k = A_k - A_{k-1}$ (where we define $A_0 = 0$), we can substitute this into our partial sum:

$$S_n = \sum_{k=1}^n (A_k - A_{k-1}) b_k.$$

Thus

$$S_n = A_n b_n + \sum_{k=1}^{n-1} A_k (b_k - b_{k+1}).$$

Now, because $|A_n| \leq M$ for all $n \in \mathbb{N}$, we have $-M b_n \leq A_n b_n \leq M b_n$. Since $\lim_{n \rightarrow \infty} b_n = 0$, both bounds approach zero. By the Squeeze Theorem (Theorem 2.10):

$$\lim_{n \rightarrow \infty} A_n b_n = 0.$$

We must now show that the series $\sum_{k=1}^{\infty} A_k (b_k - b_{k+1})$ converges. We have

$$\sum_{k=1}^{n-1} |A_k (b_k - b_{k+1})| = \sum_{k=1}^{n-1} |A_k| |b_k - b_{k+1}|,$$

and since (b_n) is decreasing, $b_k - b_{k+1} \geq 0$, so we can drop the absolute value. Using our bound $|A_k| \leq M$, we get:

$$\sum_{k=1}^{n-1} |A_k| (b_k - b_{k+1}) \leq \sum_{k=1}^{n-1} M (b_k - b_{k+1}) = M \sum_{k=1}^{n-1} (b_k - b_{k+1}) = M(b_1 - b_n).$$

Because $b_n \geq 0$, we know $M(b_1 - b_n) \leq Mb_1$. Therefore, the partial sums of the absolute series are bounded above by the constant Mb_1 . Because it is a series of non-negative terms with bounded partial sums, $\sum_{k=1}^{\infty} |A_k(b_k - b_{k+1})|$ converges. Absolute convergence implies convergence, so the series $\sum_{k=1}^{\infty} A_k(b_k - b_{k+1})$ must converge to some real number L .

Conclusion

Taking the limit of our summation by parts equation as $n \rightarrow \infty$, we have, by the Algebra of Limits:

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} A_n b_n + \lim_{n \rightarrow \infty} \sum_{k=1}^{n-1} A_k(b_k - b_{k+1}) = 0 + L = L$$

Therefore, (S_n) converges, i.e. the series $\sum_{n=1}^{\infty} a_n b_n$ converges. $\square$

Example 2.67. The series $\sum_{n=1}^{\infty} \frac{e^{in\theta}}{n}$ is convergent for all $\theta \not\equiv 0 \pmod{2\pi}$. We have

$$\left| \sum_{k=1}^n e^{ik\theta} \right| = \left| \frac{e^{i\theta}(1 - e^{in\theta})}{1 - e^{i\theta}} \right| \leq \frac{2}{|1 - e^{i\theta}|},$$

since $\theta \not\equiv 0 \pmod{2\pi}$, so $1 - e^{i\theta}$ is a non-zero constant. Also, $\frac{1}{n} \rightarrow 0$ and is decreasing, so by the Dirichlet test (Theorem 2.66), $\sum_{n=1}^{\infty} \frac{e^{in\theta}}{n}$ is convergent.

2.3 Power series

Definition 2.68 (Power series). A *power series* centred at $z_0 \in \mathbb{C}$ is a series of the form

$$P(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$$

where z is a complex variable and (a_n) is a sequence of complex coefficients.

Remark. A power series is an “infinite degree polynomial”, i.e. it is the function we get when we take the limit of finite polynomials in $\mathbb{C}$, i.e. the limit of

$$\sum_{i=0}^n a_i(z - z_0)^i = a_0 + a_1(z - z_0) + \cdots + a_n(z - z_0)^n.$$

Note that we have a *different* sequence for *each* value of $z \in \mathbb{C}$, and therefore when $P(z)$ converges, we have a different *limit* for each value of $z \in \mathbb{C}$. Power series can therefore be seen as functions $\mathbb{C} \rightarrow \mathbb{C}$. Therefore, this motivates our interest in the radius of convergence for a power series, i.e. for which values of $z \in \mathbb{C}$ does the power series converge? Since the complex plane $\mathbb{C}$ has no natural ordering (we cannot say $z_1 < z_2$), we cannot directly apply real convergence tests to $a_n(z - z_0)^n$. Instead, we apply tests to their magnitudes $|a_n(z - z_0)^n|$, so the convergence problem is in $\mathbb{R}$. We will see why this makes sense below:

Lemma 2.69. Let (a_n) be a complex sequence and let $P(z)$ be the power series associated to (a_n) , centred at $z_0 \in \mathbb{C}$. Let $w \neq z_0$ be a complex number such that $P(w)$ is convergent. Then $P(z)$ is absolutely convergent for all $z \in \mathbb{C}$ such that $|z - z_0| < |w - z_0|$.

Proof. By assumption $P(w) = \sum_{n=0}^{\infty} a_n(w - z_0)^n$ is convergent, hence $a_n(w - z_0)^n$ has to tend to 0 and $(a_n(w - z_0)^n)$ is bounded. This means that there exists $M \in \mathbb{R}_{>0}$ such that $|a_n(w - z_0)^n| \leq M$ for all $n \in \mathbb{N}$. Let $z \in \mathbb{C}$ be such that $|z - z_0| < |w - z_0|$. Then

$$|a_n(z - z_0)^n| = \left| a_n(w - z_0)^n \frac{(z - z_0)^n}{(w - z_0)^n} \right| = |a_n(w - z_0)^n| \left| \frac{z - z_0}{w - z_0} \right|^n \leq M \left| \frac{z - z_0}{w - z_0} \right|^n.$$

Since $|w - z_0| > |z - z_0|$, we have $\left| \frac{z - z_0}{w - z_0} \right| < 1$, so $P(z)$ is absolutely convergent by comparison with $M \left| \frac{z - z_0}{w - z_0} \right|^n$. $\square$

Definition 2.70 (Radius of convergence). The *radius of convergence*, R , of a power series $P(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ centred at $z_0 \in \mathbb{C}$ is defined as:

$$R := \sup \{ |z - z_0| \in \mathbb{R}_{\geq 0} : P(z) \text{ is convergent} \}.$$

Note that the supremum does not exist if the set is not bounded above. In this case we say that the power series has infinite radius of convergence, and we write $R = \infty$. Another remark is that 0 is always in the set of numbers for which the power series is convergent, hence $R \geq 0$. If $R = 0$, this means that the power series is always divergent, except for the trivial case $z = 0$.

Remark. Power series convergence regions in $\mathbb{C}$ are circular. A power series will absolutely converge for any z strictly inside the open disk defined by $|z - z_0| < R$, and it will diverge for any z strictly outside ($|z - z_0| > R$). The boundary circle $|z - z_0| = R$ is the ambiguous zone where the series might converge, diverge, or conditionally converge depending on the specific point. This is why, as mentioned earlier, why we can apply tests to the magnitudes $|a_n z^n|$, rather than having to test individual values z in the radius of convergence. However, individual values of $z \in \mathbb{C}$ such that $|z - z_0| = R$ must be tested to determine convergence or divergence on the boundary of the convergence region.

Theorem 2.71. Let $P(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ be a power series with radius of convergence R . Then

1. $P(z)$ is absolutely convergent if $|z - z_0| < R$.
2. $P(z)$ is divergent if $|z - z_0| > R$.

Proof. 1. Let $z \in \mathbb{C}$ be such that $|z - z_0| < R$. Therefore, by definition of R , there exists $r \in \mathbb{R}$ such that $|z - z_0| < r \leq R$ such that $P(r)$ is convergent. The result follows from Lemma 2.69.

2. Let $z \in \mathbb{C}$ be such that $|z - z_0| > R$, and suppose for contradiction $P(z)$ is convergent. Then there exists $r \in \mathbb{R}$ such that $|z - z_0| > r > R$, but then $P(r)$ is convergent by Lemma 2.69, contradicting the definition of R . $\square$

Theorem 2.72 (Power series ratio test). *Let $P(z) = \sum a_n(z - z_0)^n$ be a power series centred at $z_0 \in \mathbb{C}$. If the limit $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ exists (or is ∞), then the radius of convergence is given by $R = \frac{1}{L}$ (with the convention that $1/0 = \infty$ and $1/\infty = 0$).*

Proof. Case 1: $L \neq 0$. We observe that for $z \in \mathbb{C}$, we have $\left| \frac{a_{n+1}(z - z_0)^{n+1}}{a_n(z - z_0)^n} \right| \rightarrow L|z - z_0|$ as $n \rightarrow \infty$. If $L|z - z_0| < 1$ then $\sum_{n=0}^{\infty} a_n z^n$ converges by the Ratio test, and if $L|z - z_0| > 1$ then $\sum_{n=0}^{\infty} a_n z^n$ diverges, so $R = \frac{1}{L}$.

Case 2: $L = 0$. Then for any $z \in \mathbb{C}$, $\left| \frac{a_{n+1}(z - z_0)^{n+1}}{a_n(z - z_0)^n} \right| \rightarrow 0$, so $R = \infty$ (the set of values for which $P(z)$ converges is unbounded).

Case 3: $L = \infty$. Then for any $z \neq 0$, $\left| \frac{a_{n+1}(z - z_0)^{n+1}}{a_n(z - z_0)^n} \right| \rightarrow \infty$, so $(a_n(z - z_0)^n)$ is not summable for any $z \neq 0$, so $R = 0$. $\square$

Remark. Intuitively, you can think of the radius of convergence R as the outcome of an asymptotic tug-of-war between two competing forces: the growth (or decay) rate of the coefficients a_n , and the geometric growth (or decay) of $|z - z_0|^n$.

Example 2.73. 1. The power series $\sum_{n=0}^{\infty} n!(z - z_0)^n$ has convergence radius $R = 0$ because

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)!}{n!} = n+1 \rightarrow \infty.$$

2. The power series $\sum_{n=0}^{\infty} \frac{n^n}{n!}(z - z_0)^n$ has convergence radius $R = \frac{1}{e}$ because

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)^{n+1}/(n+1)!}{n^n/n!} = \frac{1}{n+1} \frac{(n+1)(n+1)^n}{n^n} = \left(1 + \frac{1}{n}\right)^n \rightarrow e.$$

3. The power series $\sum_{n=0}^{\infty} \frac{1}{n!} z^n$ has convergence radius $R = \infty$ because

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{1/(n+1)!}{1/n!} = \frac{1}{n+1} \rightarrow 0.$$

By item 3 above, the following function is well-defined:

Definition 2.74 (The complex exponential function). For any $z \in \mathbb{C}$, we define the complex exponential function as the power series:

$$\exp(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

Exercise. Prove that for any $z, w \in \mathbb{C}$, $\exp(z + w) = \exp(z) \exp(w)$. (*Hint: use the Cauchy product formula for series and the Binomial Theorem*).

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